



CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with the answer sheet. Failure to do so may result in disqualification.

TEST CODE **01234010**

FORM TP 2017084

MAY/JUNE 2017

**CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

MATHEMATICS

Paper 01 – General Proficiency

1 hour 30 minutes

10 MAY 2017 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 30 minutes to answer them.
2. In addition to this booklet, you should have an answer sheet.
3. **A list of formulae is provided on page 2 of this booklet.**
4. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
5. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

$$2a + 6a =$$

- (A) $8a$
(B) $8a^2$
(C) $12a$
(D) $12a^2$

Sample Answer



The best answer to this item is “ $8a$ ”, so (A) has been shaded.

6. If you want to change your answer, erase it completely before you fill in your new choice.
7. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
8. You may do any rough work in this booklet.
9. Calculators and mathematical tables are NOT allowed for this paper.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF FORMULAE

Volume of a prism	$V = Ah$ where A is the area of the cross-section and h is the perpendicular length.
Volume of cylinder	$V = \pi r^2 h$ where r is the radius of the base and h is the perpendicular height.
Volume of a right pyramid	$V = \frac{1}{3} Ah$ where A is the area of the base and h is the perpendicular height.
Circumference of a circle	$C = 2\pi r$ where r is the radius of the circle.

Arc length	$S = \frac{\theta}{360} \times 2\pi r$ where θ is the angle subtended by the arc, measured in degrees
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Area of a circle	$A = \pi r^2$ where r is the radius of the circle.
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Area of a sector of a circle	$A = \frac{\theta}{360} \times \pi r^2$ where θ is the angle in the sector measured in degrees.
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Area of trapezium	$A = \frac{1}{2} (a + b) h$ where a and b are the lengths of the parallel sides and h is the perpendicular distance between the parallel sides.
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Roots of quadratic equations	If $ax^2 + bx + c = 0$,
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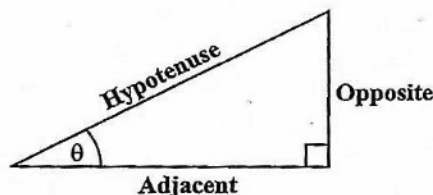
$$\text{then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometric ratios

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$



Area of triangle

Area of $\triangle ABC = \frac{1}{2}bh$ where b is the length of the base and h is the perpendicular height

$$\text{Area of } \triangle ABC = \frac{1}{2}ab \sin C$$

$$\text{Area of } \triangle ABC = \sqrt{s(s-a)(s-b)(s-c)}$$

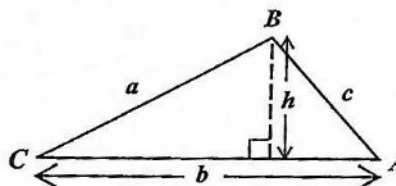
$$\text{where } s = \frac{a+b+c}{2}$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Sine rule

Cosine rule

$$a^2 = b^2 + c^2 - 2bc \cos A$$



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1. $(-3)^2 + (-2)^2$ is equal to
- (A) -13
(B) -10
(C) 13
(D) 25
2. What number when added to $1\frac{1}{3}$ gives 2?
- (A) $\frac{1}{3}$
(B) $\frac{2}{3}$
(C) 1
(D) 3
3. Ann and Betty shared a sum of money in the ratio 2:3. If Ann received \$60 less than Betty, what was the total amount of money shared?
- (A) \$ 72
(B) \$ 80
(C) \$180
(D) \$300
4. If $235 \times 48.7 = 11\,444.5$, then $23.5 \times 0.487 =$
- (A) 11.4445
(B) 114.445
(C) 1 144.45
(D) 11 444.4
5. A test was marked out of 80. A boy scored 60% of the marks on the test. How many marks did he score?
- (A) 20
(B) 48
(C) 60
(D) 75
6. If 30% of a number is 45, what is $\frac{4}{5}$ of the number?
- (A) 36
(B) 120
(C) 150
(D) 180
7. What is the value of the digit 2 in the number 48.621?
- (A) $\frac{2}{100}$
(B) $\frac{2}{10}$
(C) 2
(D) 200
8. The highest common factor (HCF) of 12, 15 and 60 is
- (A) 1
(B) 3
(C) 12
(D) 60
9. The next term in the sequence 1, 6, 13, 22, 33 is
- (A) 44
(B) 45
(C) 46
(D) 52
10. By the distributive law, $49 \times 17 + 49 \times 3 =$
- (A) $52 + 66$
(B) 52×66
(C) $49 + 20$
(D) 49×20

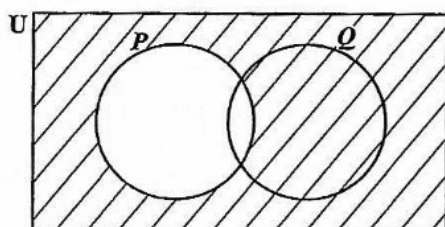
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11. If $P = \{2, 3, 5, 7\}$,
 $Q = \{2, 3, 6\}$ and
 $S = \{2, 4, 5\}$

Then $P \cap Q \cap S =$

- (A) $\{2\}$
 (B) $\{2, 3\}$
 (C) $\{2, 3, 5\}$
 (D) $\{2, 3, 4, 5, 6, 7\}$

Item 12 refers to the following Venn diagram.



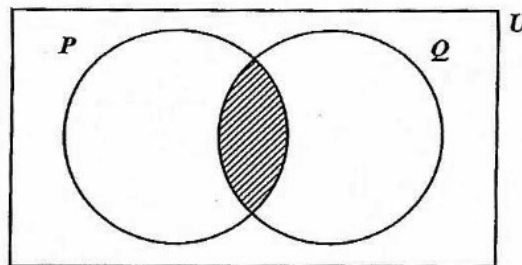
12. In the Venn diagram, the shaded area represents

- (A) P'
 (B) $(P \cup Q)'$
 (C) $Q \cup P'$
 (D) $Q \cap P'$

13. In a class of 32 students, 17 study Music and 20 study Art. What is the LEAST number of students who study BOTH Music and Art?

- (A) 3
 (B) 5
 (C) 12
 (D) 15

Item 14 refers to the following Venn diagram.



14. If $P = \{\text{Factors of 6}\}$ and $Q = \{\text{Factors of 4}\}$, then the shaded region represents

- (A) $\{ \}$
 (B) $\{1, 2\}$
 (C) $\{4, 6, 8, \dots\}$
 (D) $\{12, 24, 36, \dots\}$

15. A dress which costs \$180 is being sold at a discount of 10%. The amount of the discount is

- (A) \$ 1.80
 (B) \$ 10.00
 (C) \$ 18.00
 (D) \$170.00

16. The value of a plot of land is \$18 000. Land tax is charged at the rate of \$0.70 per \$100 value. What is the TOTAL amount of tax paid for the land?

- (A) \$110.00
 (B) \$126.00
 (C) \$180.70
 (D) \$257.15

17. At the end of any year a car is worth 5% less than what it was worth at the beginning of the year. If a car is worth \$9 500 in December 2016, then its value in January 2016 was

- (A) \$ 9 995
 (B) \$10 000
 (C) \$10 025
 (D) \$12 000

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18. A man bought a mobile phone for \$800 and sold it for \$1 000. What was his profit as a percentage of the cost price?
- (A) 5%
(B) 15%
(C) 20%
(D) 25%
19. A man's annual salary is \$45 000. His tax-free allowances total \$13 000. He pays tax of 35% on his taxable income. The tax payable is
- (A) \$ 4 550
(B) \$11 200
(C) \$15 750
(D) \$20 300
20. The simple interest on a loan of \$6 000 for 3 years was \$900. What was the rate of interest per annum?
- (A) 2%
(B) 3%
(C) 5%
(D) 18%
21. A television set costs \$350 cash. When bought on hire-purchase, a deposit of \$35 is required, followed by 12 monthly payments of \$30 each. How much is saved by paying cash?
- (A) \$10
(B) \$25
(C) \$40
(D) \$45
22. How much does a customer pay for an article marked at \$50 if a sales tax of 6% is charged?
- (A) \$44
(B) \$47
(C) \$53
(D) \$56
23. Seven times the product of two numbers, a and b , may be written as
- (A) $7ab$
(B) $49ab$
(C) $7a + b$
(D) $7(a + b)$
24. If x is an integer that satisfies the inequality $4 < 2x \leq 6$, then
- (A) $2 < x \leq 3$
(B) $-2 < x \leq 3$
(C) $-3 < x \leq 2$
(D) $-3 \leq x < -2$
25. The coefficient of x in the expansion of $(x + 3)(x - 1)$ is
- (A) -3
(B) -2
(C) 2
(D) 4
26. $3x^2 \times 2x^3 =$
- (A) $6x$
(B) $6x^5$
(C) $6x^6$
(D) $72x^5$

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27. $P = \frac{m^3}{2-m}$. When $m = -2$, the value of P is

(A) -2
(B) $-\frac{6}{4}$
(C) $\frac{6}{4}$
(D) 2

28. If $a * b = \frac{a}{b} - 1$, then $7 * 28 =$

(A) $-\frac{3}{4}$
(B) $\frac{1}{4}$
(C) 3
(D) 4

29. If $5(2x - 1) = 35$, then $x =$

(A) -4
(B) $\frac{1}{4}$
(C) 3
(D) 4

30. John has x marbles and Max has twice as many marbles as John. Max gives John 5 of his marbles. How many marbles does John now have?

(A) $x + 5$
(B) $x - 5$
(C) $2x - 5$
(D) $2x + 5$

31. When 8 is subtracted from a certain number and the result is multiplied by 3 the answer is 21. What is the original number?

(A) 1
(B) 3
(C) 10
(D) 15

32. How many kilograms are there in one tonne?

(A) 10
(B) 100
(C) $1\ 000$
(D) $10\ 000$

33. The volume, in cm^3 , of a cube of edge 3 cm is

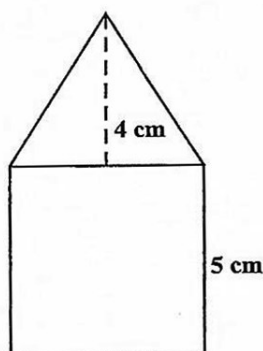
(A) 9
(B) 18
(C) 27
(D) 54

34. A car travels 80 kilometres in $2\frac{1}{2}$ hours. What is its speed in kilometres per hour?

(A) 6
(B) 32
(C) 82.5
(D) 200

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Item 35 refers to the following figure which shows a triangle resting on a square.



35. The length of one side of the square is 5 cm and the height of the triangle is 4 cm. What is the TOTAL area of the figure, in cm^2 ?
- (A) 35
(B) 45
(C) 50
(D) 100
36. The area of a rectangle is 53.6 cm^2 . If the length is multiplied by four and the width is halved, the area would then be
- (A) 26.8 cm^2
(B) 53.6 cm^2
(C) 107.2 cm^2
(D) 214.4 cm^2
37. A man leaves home at 22:15 hours and reaches his destination at 04:00 hours on the following day in the same time zone. How many hours did the journey take?
- (A) 5
(B) $5\frac{3}{4}$
(C) 6
(D) $6\frac{1}{4}$

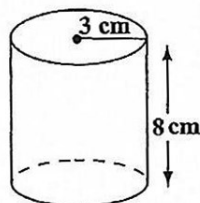
38. The distance around a lake is 8 km. On a map, this distance around the lake is represented by a length of 2 cm. The scale on the map is

(A) 1 : 40
(B) 1 : 2 000
(C) 1 : 200 000
(D) 1 : 400 000

39. The perimeter of a square is 56 cm. What is its area, in cm^2 ?

(A) 28
(B) 78
(C) 169
(D) 196

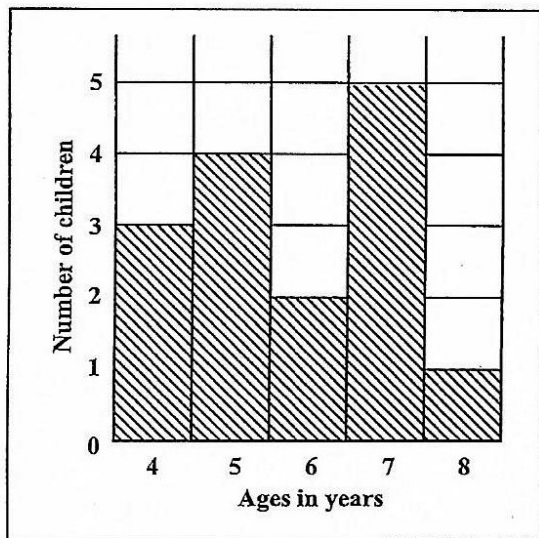
Item 40 refers to the following diagram, not drawn to scale, of a cylinder of radius 3 cm and height 8 cm.



40. The volume of the cylinder is
- (A) $12 \pi \text{ cm}^3$
(B) $48 \pi \text{ cm}^3$
(C) $72 \pi \text{ cm}^3$
(D) $192 \pi \text{ cm}^3$

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Items 41–42 refer to the following histogram which shows the number of children aged 4, 5, 6, 7 and 8 who took part in a survey.



41. What was the modal age?

- (A) 5
- (B) 6
- (C) 7
- (D) 8

42. How many children took part in the survey?

- (A) 5
- (B) 15
- (C) 75
- (D) 87

43. The following table shows the frequency of scores obtained by students in a test.

Scores	2	3	5	6	8	10
No. of Students	8	4	6	3	9	2

The range of the scores is

- (A) 2
- (B) 7
- (C) 8
- (D) 10

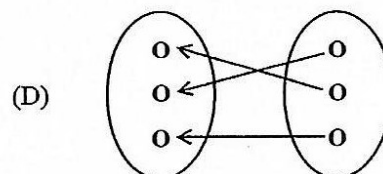
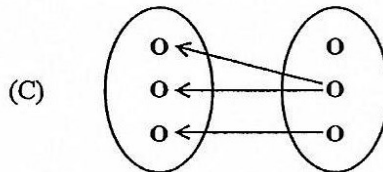
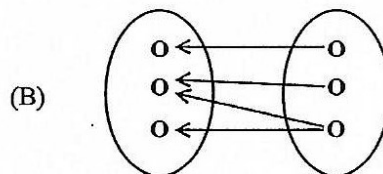
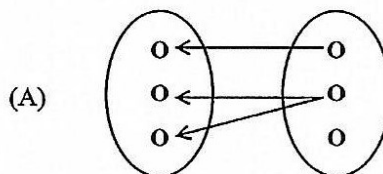
44. If the mean of the four numbers 4, 8, $2x$ and 12 is 10, then x is

- (A) 8
- (B) 10
- (C) 12
- (D) 16

45. Of 120 students writing an exam, 100 are expected to pass. The estimated probability that a randomly selected student will FAIL the exam is

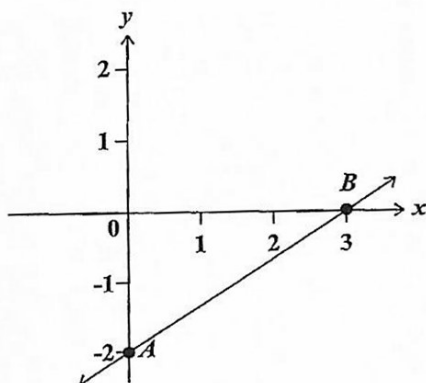
- (A) $\frac{1}{6}$
- (B) $\frac{1}{5}$
- (C) $\frac{1}{2}$
- (D) $\frac{5}{6}$

46. Which of the following diagrams BEST illustrates a function?



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Item 47 refers to the following graph of a straight line.



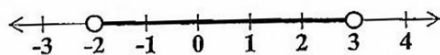
47. The straight line AB cuts the y axis at

(A) $(0, 3)$
 (B) $(0, 2)$
 (C) $(3, -2)$
 (D) $(0, -2)$

48. If $h(x) = \frac{3x-2}{5}$, then $h(-6) =$

(A) -4
 (B) $-\frac{16}{5}$
 (C) $\frac{16}{5}$
 (D) 4

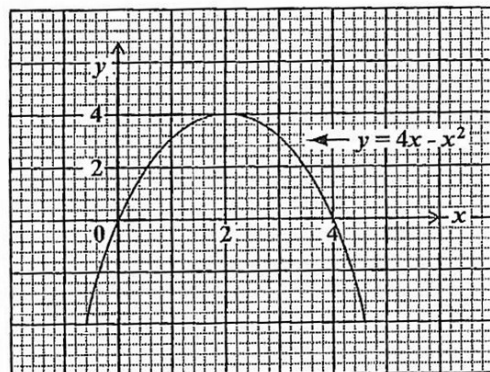
Item 49 refers to the following number line which shows the solution of an inequality.



49. The inequality above is defined by

(A) $-2 \leq x < 3$
 (B) $-2 < x \leq 3$
 (C) $-2 \leq x \leq 3$
 (D) $-2 < x < 3$

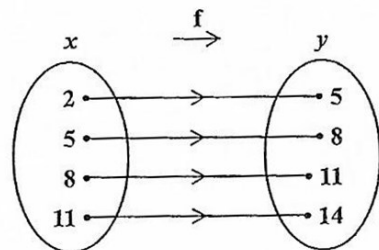
Item 50 refers to the following graph of a function.



50. The values of x for which $y = 4x - x^2$ intersects $y = 0$ are

(A) $x = 0$ and $x = 4$
 (B) $x = 0$ and $x = 2$
 (C) $x = 0$ and $x = -4$
 (D) $x = 2$ and $x = 4$

Item 51 refers to the following arrow diagram which shows a function.



51. Which of the following BEST describes the function?

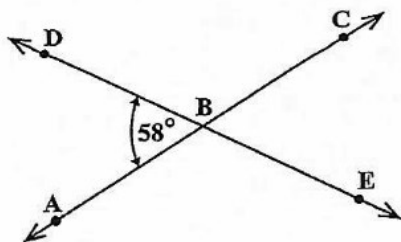
(A) $f(x) = x$
 (B) $x = f(x) + 3$
 (C) $f(x) = y + 3$
 (D) $f(x) = x + 3$

52. If each exterior angle of a regular polygon is 60° , then the polygon is a

(A) quadrilateral
 (B) pentagon
 (C) hexagon
 (D) triangle

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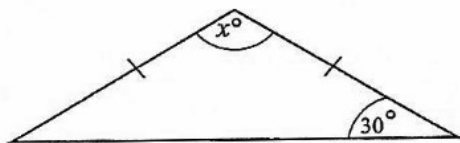
Item 53 refers to the following diagram of a pair of intersecting lines.



53. The measure of angle ABE is

- (A) 58°
- (B) 122°
- (C) 142°
- (D) 302°

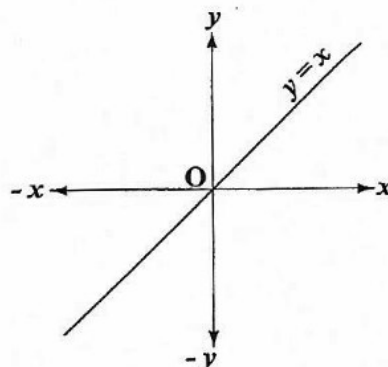
Item 54 refers to the following diagram of an isosceles triangle.



54. In the triangle, the value of x is

- (A) 30°
- (B) 60°
- (C) 120°
- (D) 150°

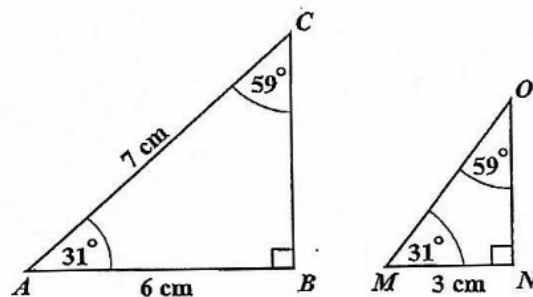
Item 55 refers to the following diagram of the straight line $y = x$.



55. What is the image of the line $y = x$ when it is rotated anticlockwise about O through an angle of 90° ?

- (A) $y = 0$
- (B) $x = 0$
- (C) $y = x$
- (D) $y = -x$

Item 56 refers to the following pair of similar triangles.

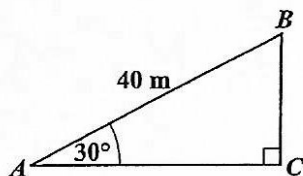


56. The length of MO , in centimetres, is

- (A) 2.5
- (B) 3.0
- (C) 3.5
- (D) 7.0

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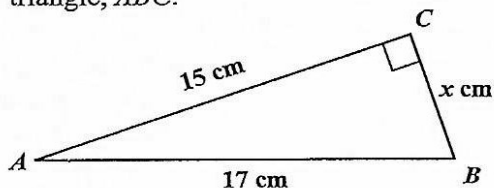
Item 57 refers to the following right-angled triangle, ABC .



57. In the triangle, **not drawn to scale**, angle $BAC = 30^\circ$ and $AB = 40$ m. The length BC , in metres, is

- (A) $40 \sin 30^\circ$
- (B) $40 \tan 30^\circ$
- (C) $40 \sin 60^\circ$
- (D) $40 \tan 60^\circ$

Item 58 refers to the following right-angled triangle, ABC .



58. In the triangle ABC , **not drawn to scale**, AB is 17 cm, AC is 15 cm and BC is x cm. The value of x , in centimetres, is

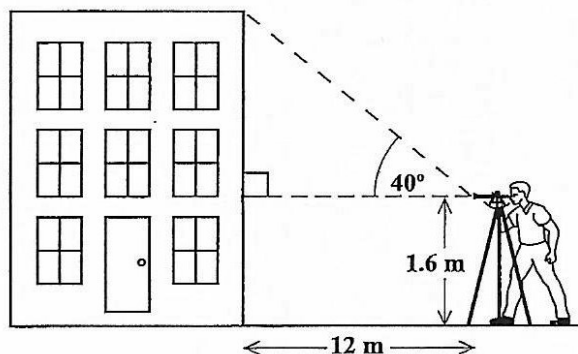
- (A) 7
- (B) 8
- (C) 9
- (D) 12

59. In a triangle, PQR , angle $P = x^\circ$ and angle $Q = 2x^\circ$. What is the size of angle R ?

- (A) 60°
- (B) 45°
- (C) $\left(\frac{180}{3x}\right)^\circ$
- (D) $(180^\circ - 3x^\circ)$

Item 60 refers to the following diagram of a building.

A surveyor sets up his instruments 12 metres from the foot of a building, and records the angle of elevation of the top of the building.



60. An estimate of the height of the building is obtained by calculating

- (A) $12 \tan 40^\circ$
- (B) $12 \sin 40^\circ$
- (C) $1.6 + 12 \cos 40^\circ$
- (D) $1.6 + 12 \tan 40^\circ$

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.